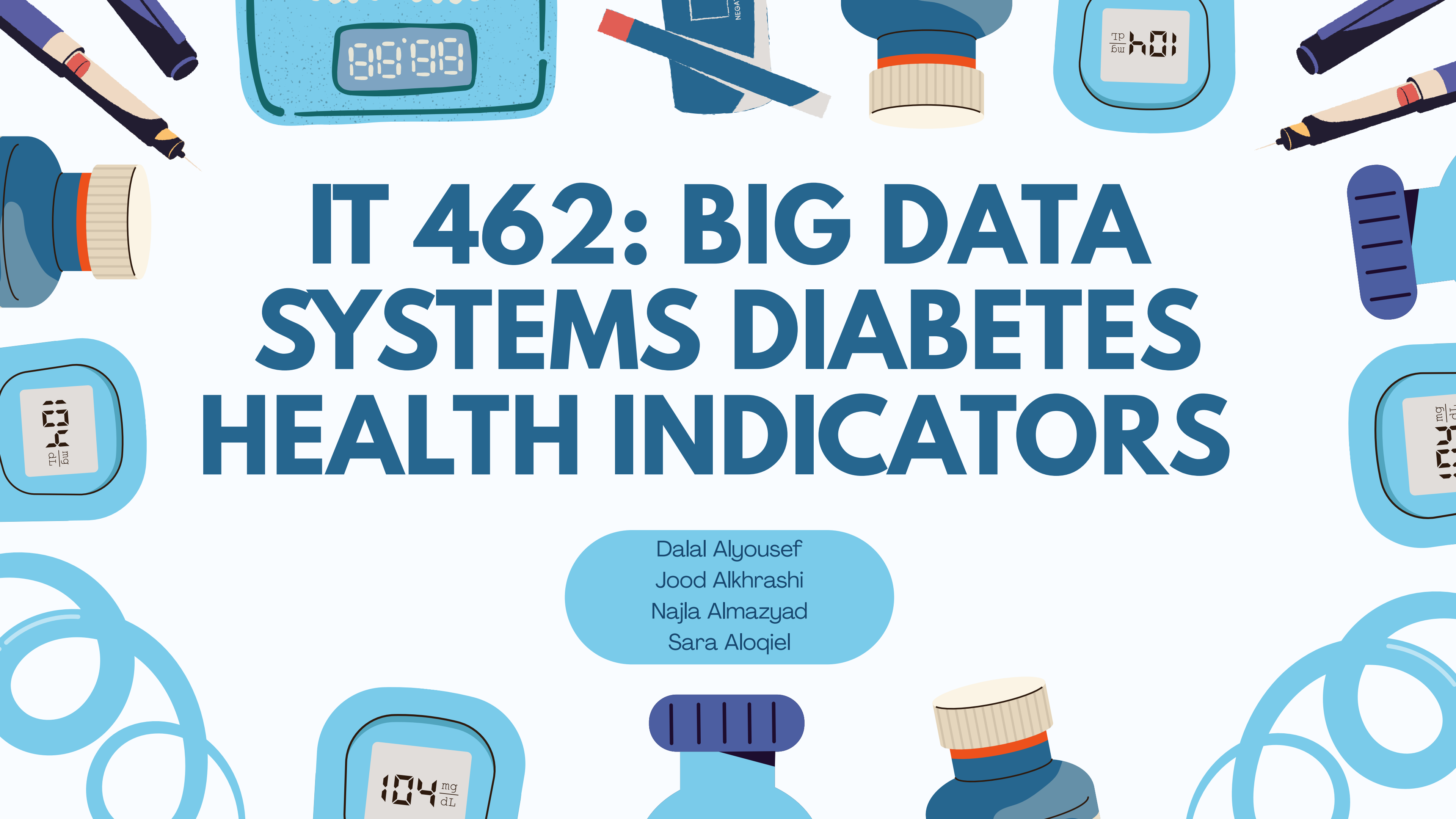


The background features a decorative border of various medical and technology-related icons. These include syringes, digital blood pressure monitors, insulin pens, and pill bottles. The icons are rendered in a flat, illustrative style with a color palette of blues, purples, and light blues.

IT 462: BIG DATA SYSTEMS DIABETES HEALTH INDICATORS

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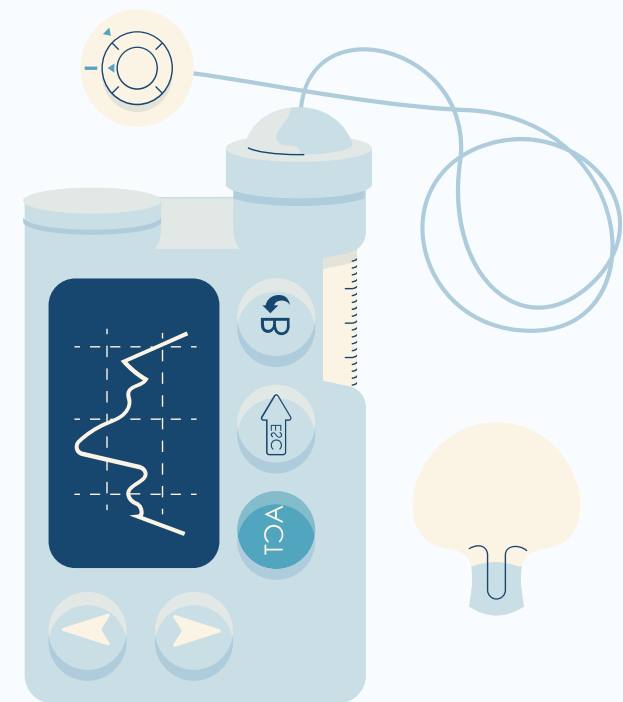
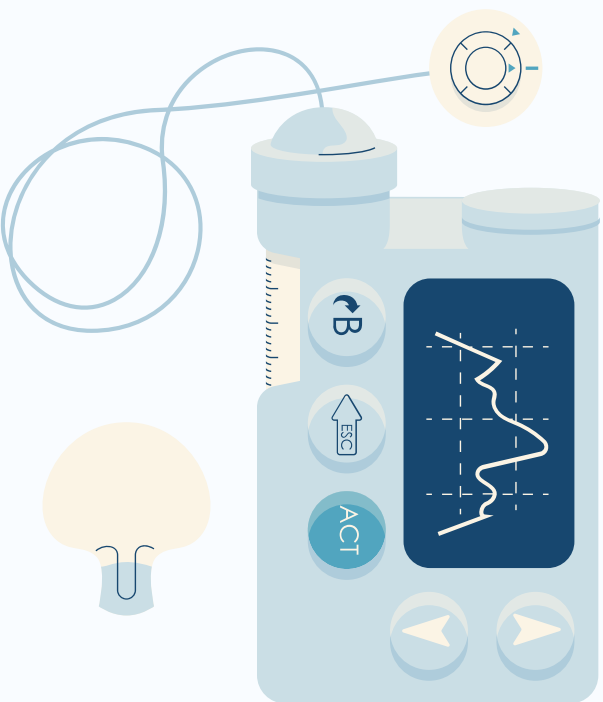
PROBLEM & DATASET

The Problem Millions remain undiagnosed with diabetes until serious complications arise. Can we predict diabetes risk using only survey data without any lab tests, nor clinical visits?

The Dataset

- **Source:** CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015
- **229,500 records** • 22 variables • ~22 MB CSV
- **Target:** Diabetes_binary (0 = No Diabetes, 1 = Prediabetes, 2 = Diabetes)

Why It Matters Early identification using lifestyle and demographic data enables cheaper, wider screening, especially in underserved communities.



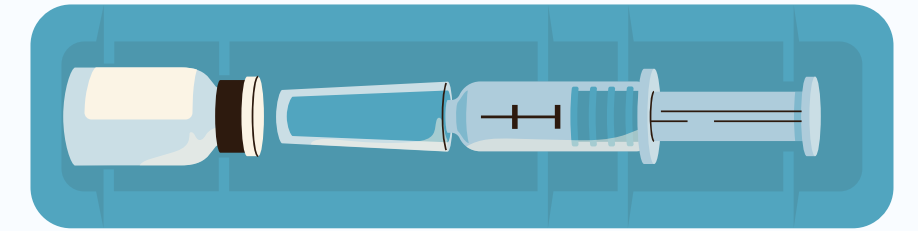
PREPROCESSING & EXPLORATION

Main Challenges

- **Class imbalance:** majority of records are non-diabetic
- **Mixed variable types:** binary, ordinal, and continuous features

Interesting Patterns Found Early

- **Diabetes rate rises sharply with age:** ~7% at age 18–24 → over 40% at 80+
- **Over 77%** of the data population is **overweight** or **obese**
- **22%** of respondents self-report **Fair** or **Poor** general health



RDD HIGHLIGHTS

Key RDD Operations Used

map · filter · reduceByKey · mapValues · sortBy · sortByKey

Insight 1: Age & Diabetes Diabetes rate climbs steadily with age group. Age is the single clearest demographic predictor.

Insight 2: High-Risk Clustering 14,823 individuals (6.46%) simultaneously have high BP, smoking history, physical inactivity, and poor self-rated health risk factors cluster together, not appear in isolation.

Insight 3: BMI Distribution 41% of the dataset falls in the Obese category ($BMI \geq 30$). Average BMI: 31.86 for diabetic vs 27.43 for non-diabetic individuals.



RDD HIGHLIGHTS



Diabetes Across Age Groups

Value
18 - 24
25 - 29
30 - 34
35 - 39
40 - 44
45 - 49
50 - 54
55 - 59
60 - 64
65 - 69
70 - 74
75 - 79
80 - 99

2.3 Diabetes Distribution by Age Group				
Age Group Diabetes Cases Total Diabetes Rate (%)				
Age 1	177	5502	3.22%	
Age 2	332	7057	4.70%	
Age 3	697	9988	6.98%	
Age 4	1390	12220	11.37%	
Age 5	2256	14026	16.08%	
Age 6	3787	17275	21.92%	
Age 7	6556	23114	28.36%	
Age 8	9012	27260	33.06%	
Age 9	12060	29708	40.60%	
Age 10	13639	29138	46.81%	
Age 11	10775	22027	48.92%	
Age 12	7204	15381	46.84%	
Age 13	6850	16804	40.76%	

SQL HIGHLIGHTS

5 Analytical Queries Run with Spark SQL

Query	Key Finding
Diabetes Rate by Age Group	Rate rises from 1.80% (Age 1) to 25.82% (Age 11)
Avg BMI by Diabetes Status	Non-diabetic: 27.96 Prediabetes/Diabetic: >30
High-Risk Individual Count	3,823 with 4+ combined risk factors
Top Age Brackets by Prevalence	Age codes 10–13 consistently highest
BMI Category Distribution	Obese: 84,661 · Overweight: 82,859

SQL confirmed what RDD exploration suggested: age and BMI are the dominant structured predictors of diabetes risk in this dataset.

MACHINE LEARNING: SETUP

Task: Multi-class classification

- Class 0: No Diabetes
- Class 1: Prediabetes
- Class 2: Diabetes

Pipeline Stages Raw Features → StringIndexer → VectorAssembler → RandomForestClassifier → Predictions

Why Random Forest?

- Handles mixed variable types well
- Robust to noise and nonlinear relationships
- Produces feature importance scores for interpretability

Hyperparameters

Trees: 50 · Max Depth: 10 · Seed: 42

Split: 70% train / 30% test



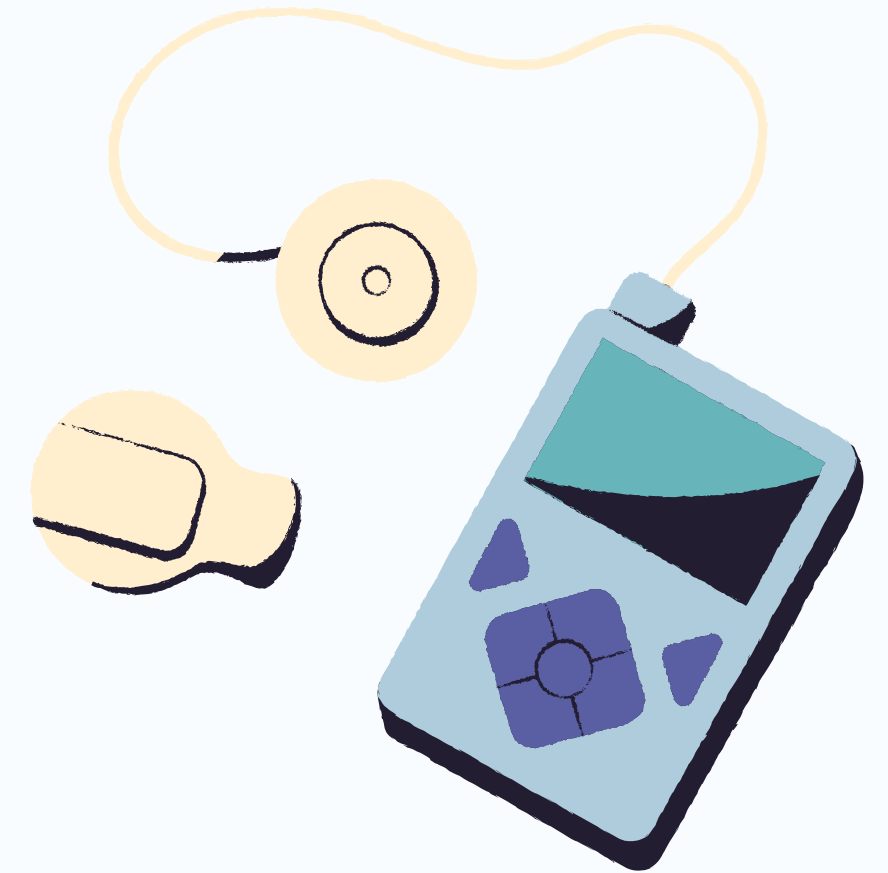
MACHINE LEARNING: RESULTS

Model	Accuracy	F1-Score	Baseline Accuracy
Random Forest	83.49%	78.28%	82.82%

What this means: The model beats the baseline, but the gap is narrow. Class imbalance keeps inflating the baseline. Accuracy alone is misleading here; F1-score of 78.28% better reflects real predictive power across all three classes.

Top Predictors (Feature Importance)

1. General Health (GenHlth)
2. Blood Pressure (HighBP)
3. Body Mass Index (BMI)
4. Age
5. Income





THANK YOU

**ANY
QUESTIONS?**

